

Tournament Predictions

```
In [1]: #setup
from rugbyModels import *
np.random.seed(1)
import warnings
warnings.filterwarnings("ignore")

data= pd.read_csv("final_data_1.csv")
data["date"] = pd.to_datetime(data["date"])

#think its reasoable to combine these for models without dgs - shouldnt overwrit
data["home_penalties"] = data["home_penalties"]+data["home_drop_goals"]
data["away_penalties"] = data["away_penalties"]+data["away_drop_goals"]
```

```
In [2]: ncPred = PoissonModel().fit(data) # fit model
NHtable,SHtable = ncPred.simulate_group() # simulate group stage
```

Initializing NUTS using jitter+adapt_diag...
Multiprocess sampling (4 chains in 4 jobs)
NUTS: [home, sd_att, sd_def, att, def_, int_try, sd_dis_clean, sd_dis_force, dis_clean, dis_force, int_pen, p_conv]
Output()

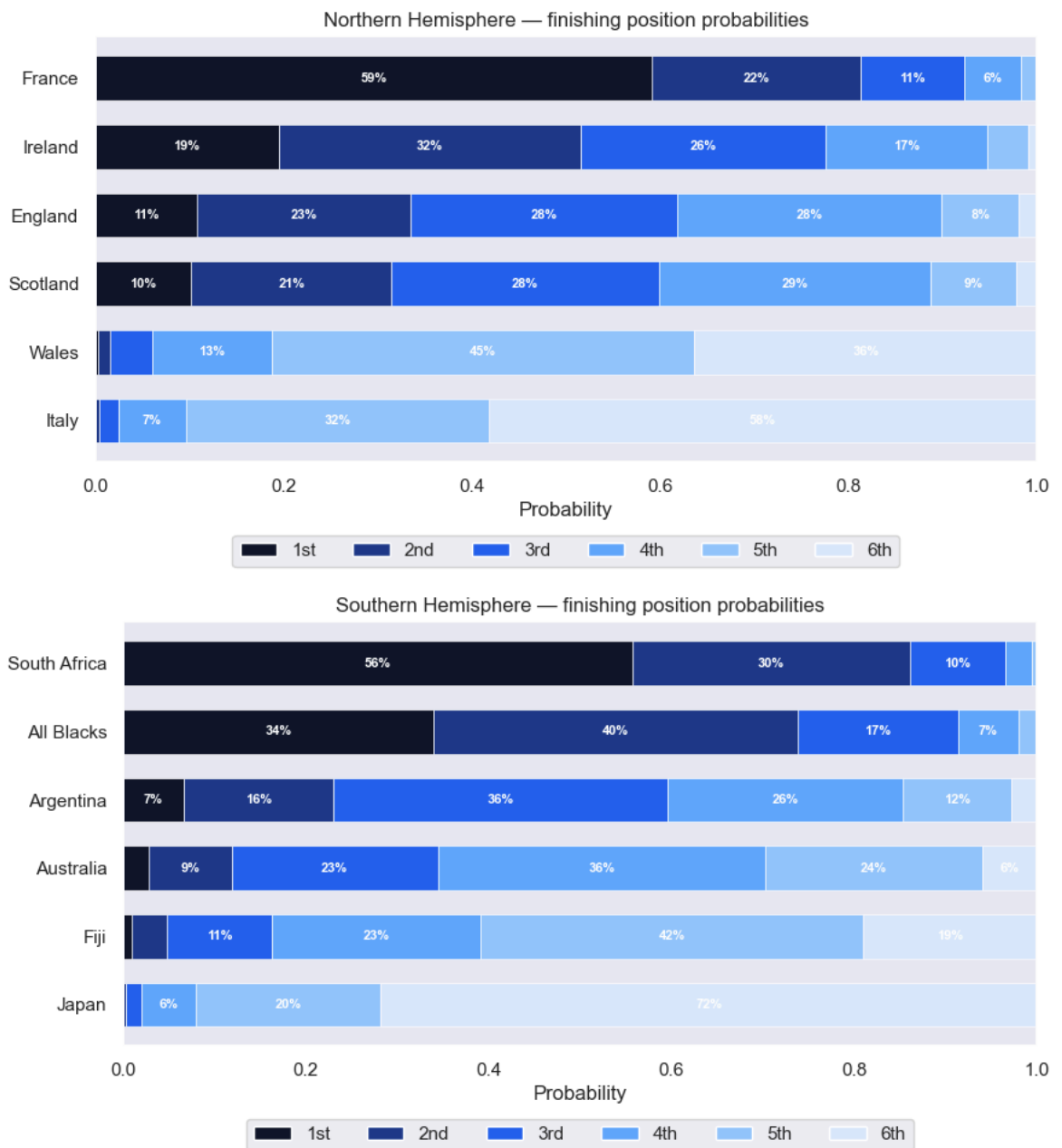
Sampling 4 chains for 1_500 tune and 2_500 draw iterations (6_000 + 10_000 draws total) took 236 seconds.
The rhat statistic is larger than 1.01 for some parameters. This indicates problems during sampling. See <https://arxiv.org/abs/1903.08008> for details
The effective sample size per chain is smaller than 100 for some parameters. A higher number is needed for reliable rhat and ess computation. See <https://arxiv.org/abs/1903.08008> for details

=== Northern Hemisphere ===

	mean_pts	p_1st	p_top2	p_top3	p_top4	p_top5
France	23.639	0.591	0.813	0.923	0.983	0.998
Ireland	19.560	0.195	0.516	0.775	0.947	0.991
England	17.757	0.108	0.335	0.618	0.899	0.980
Scotland	17.536	0.102	0.315	0.598	0.887	0.977
Wales	9.724	0.004	0.016	0.061	0.188	0.636
Italy	7.674	0.001	0.005	0.025	0.096	0.418

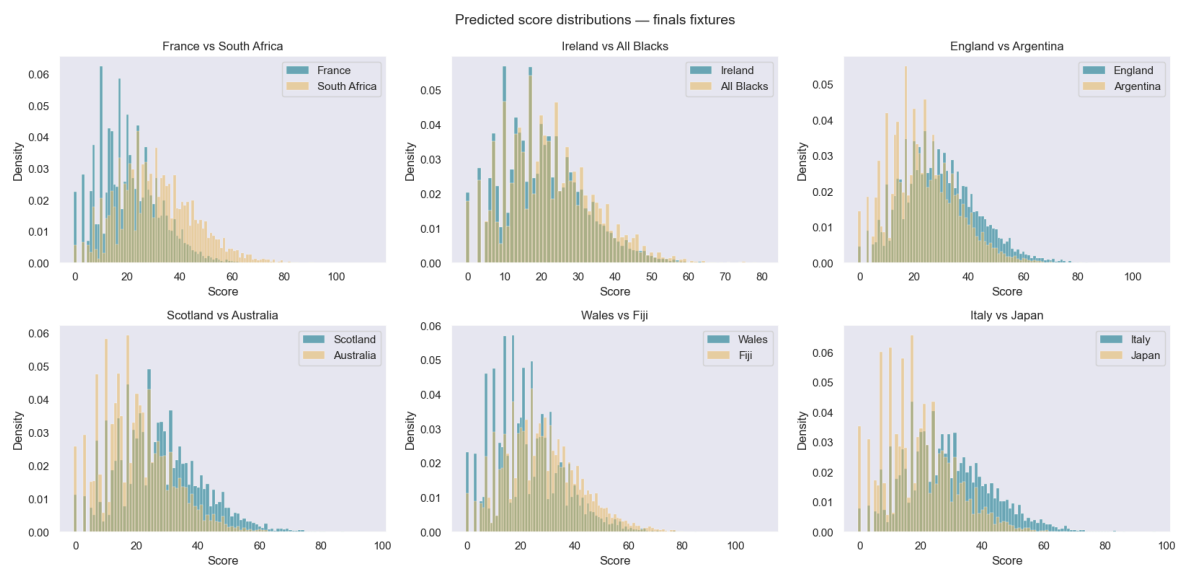
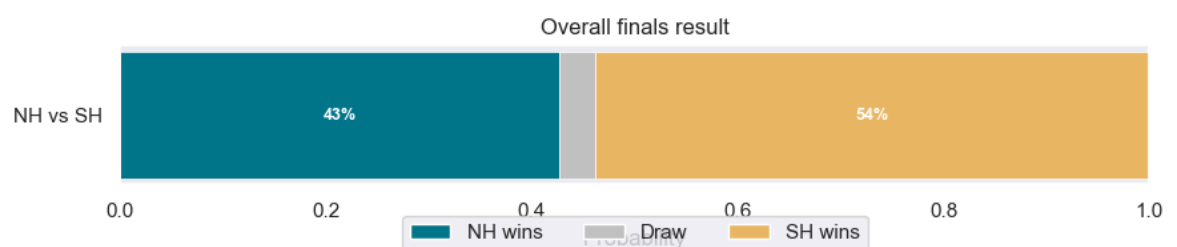
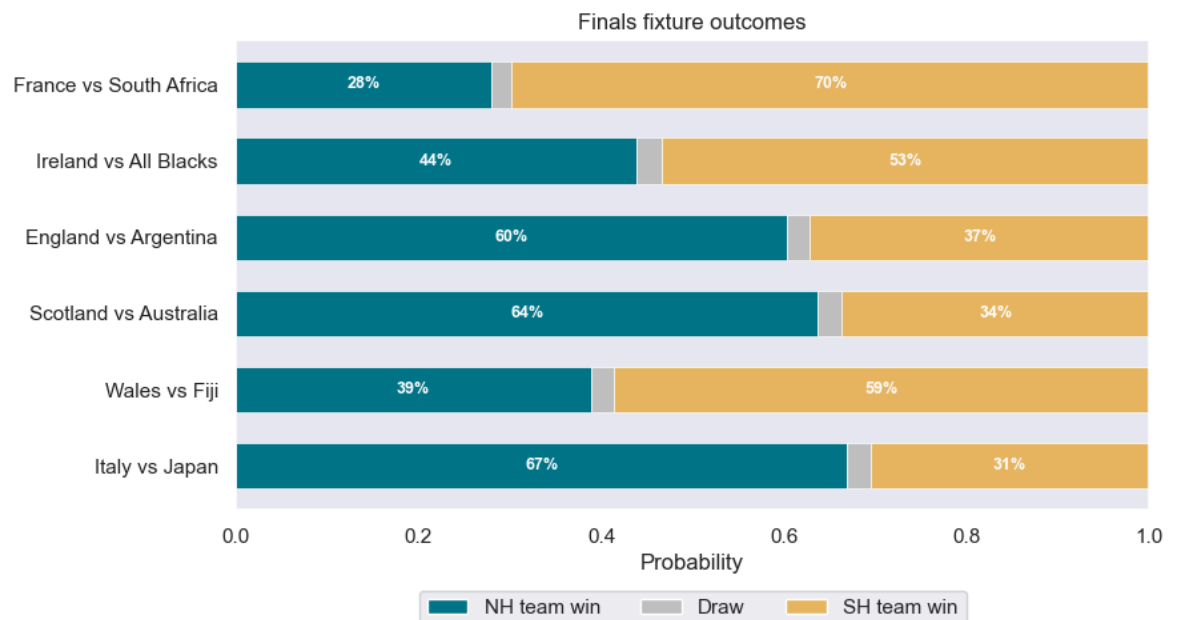
=== Southern Hemisphere ===

	mean_pts	p_1st	p_top2	p_top3	p_top4	p_top5
South Africa	22.808	0.557	0.861	0.965	0.994	1.000
All Blacks	21.008	0.339	0.739	0.913	0.979	0.998
Argentina	15.550	0.066	0.231	0.595	0.853	0.972
Australia	13.084	0.028	0.119	0.345	0.702	0.940
Fiji	9.910	0.009	0.048	0.163	0.391	0.810
Japan	4.977	0.000	0.003	0.020	0.080	0.282



And if we assume these group placings, how will the finals go?

```
In [3]: _ = ncPred.simulate_finals(NHtable, SHtable)
```



=== Finals fixtures ===

France vs South Africa: $P(\text{North})=0.280$ $P(\text{Draw})=0.022$ $P(\text{South})=0.698$

Ireland vs All Blacks: $P(\text{North})=0.438$ $P(\text{Draw})=0.027$ $P(\text{South})=0.534$

England vs Argentina: $P(\text{North})=0.603$ $P(\text{Draw})=0.025$ $P(\text{South})=0.372$

Scotland vs Australia: $P(\text{North})=0.637$ $P(\text{Draw})=0.026$ $P(\text{South})=0.337$

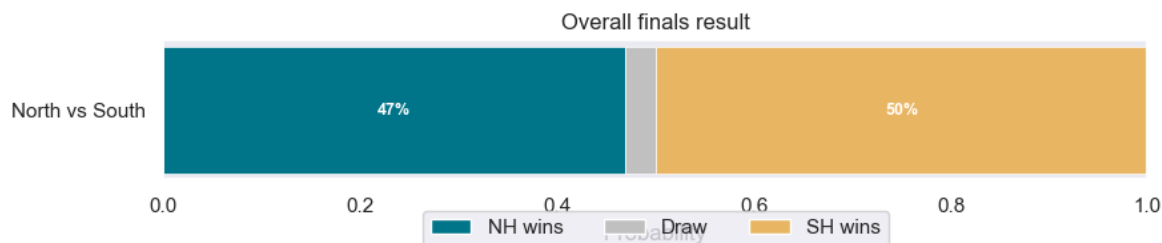
Wales vs Fiji: $P(\text{North})=0.388$ $P(\text{Draw})=0.026$ $P(\text{South})=0.586$

Italy vs Japan: $P(\text{North})=0.669$ $P(\text{Draw})=0.025$ $P(\text{South})=0.305$

Overall: $P(\text{NH wins}) = 0.426$ $P(\text{Tie}) = 0.036$ $P(\text{SH wins}) = 0.538$

And if we don't assume these placings and factor uncertainty over group stage positions into finals calculations?

```
In [4]: _ = ncPred.simulate_finals_from_scratch()
```



=== Finals ===

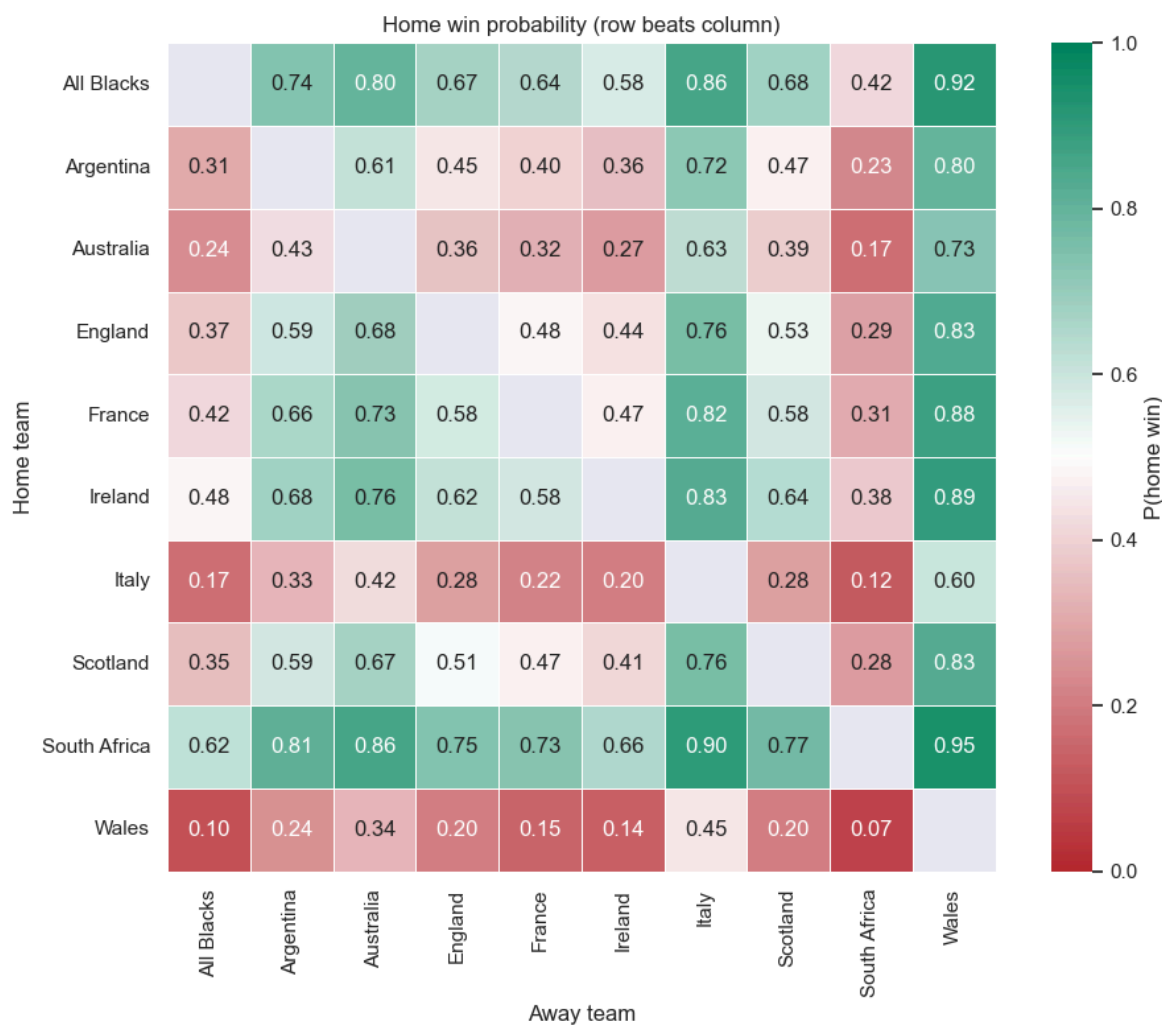
Overall: $P(\text{North}) = 0.469$ $P(\text{Tie}) = 0.032$ $P(\text{South}) = 0.500$

Looking close!

Pairwise win probabilitie

The probability of each team winning a home fixture against each other team.

```
In [5]: ncPred.plot_pairwise_win_probs()
```



Out[5]:

	All Blacks	Argentina	Australia	England	France	Ireland	Italy	Scotland	South Africa
All Blacks	NaN	0.7364	0.8007	0.6687	0.6449	0.5760	0.8576	0.6818	0.4000
Argentina	0.3080	NaN	0.6132	0.4463	0.3984	0.3580	0.7187	0.4651	0.2000
Australia	0.2404	0.4295	NaN	0.3626	0.3208	0.2746	0.6282	0.3886	0.1000
England	0.3745	0.5929	0.6836	NaN	0.4835	0.4358	0.7637	0.5346	0.2000
France	0.4155	0.6565	0.7341	0.5789	NaN	0.4679	0.8174	0.5829	0.3000
Ireland	0.4800	0.6805	0.7619	0.6161	0.5821	NaN	0.8286	0.6402	0.3000
Italy	0.1708	0.3327	0.4237	0.2801	0.2189	0.2030	NaN	0.2832	0.1000
Scotland	0.3523	0.5872	0.6728	0.5148	0.4735	0.4089	0.7638	NaN	0.2000
South Africa	0.6180	0.8110	0.8605	0.7458	0.7323	0.6562	0.9044	0.7696	NaN
Wales	0.1045	0.2448	0.3351	0.2013	0.1473	0.1373	0.4463	0.2016	0.0000

Interpretability

What is driving each teams success/failure?

```
In [6]: ncPred.plot_params()
```

